

Immunology and Sereology LAB

LAB (1)

Immune system:

Innate

- start at born
- No need for exposing to Forigen substances
- No memory
- Not specific
- Rapid

example: skin, mucus, saliva,
macrophages, Acidic PH
of the stomach.

Adaptive (aquired)

- Need for exposing to Forigen substances
- Memory
- Specific
- slow

example:

cellular

NK, T-lymphocyte

humoral

B-lymphocyte
↓
antibody

Antigen: Forigen substance that could or couldn't enhance immune system.

Immunogen: Forigen substance that must enhance immune system.

Antibody: Protein Produced by B-lymphocyte called immunoglobulin (IgM, IgG, IgA, IgE, IgD)

- Characters of antigens that enhance immune system: it must be:
 - Protein with high molecular mass
 - must be sterile (free of microorganism)
 - must be pure (one type)

• have a suitable PH

• suitable dose

High dose

suitable dose

low dose

- tolerance

- strong immune response.

- no immune response.

- no immune response

• Route of admission of the Porigen substance:

- mouth and nose

- Intradermis I.D "inside skin"

- Intramuscular I.M (inside muscle)

- Subcutaneous S.C "under skin"

- Intraperitoneal I.P

- Intravenous I.V "side vein"

done without adjuvant.

• Types of antigen according to rxn with antibody:

Soluble Ag

Particle Ag

medium

large

soluble Ag + soluble Ab = Apt

Particle Ag + soluble Ab = Agglutination

Bovine serum Albumin

cell wall & cell membrane

Add adjuvant

Not add adjuvant

Lab (2)

Adjuvant: substance that add to soluble Ag to enhance immune system by two ways:

- slow released Ag to tissue.

- slow absorb of Ag by tissue.

(F.A)
Freud Adjuvant

Complete F.A

C.F.A

oil + weaken mycobacterium
at first injection

Incomplete F.A

I.F.A

oil only
at 2nd, 3rd, 4th, ...

* 3 days between the first and second injection

IgM

~~IgM~~
IgG

Animals

① rat

② rabbit

③ hamster

④ chicken

⑤ mice

↓
- bulb C white mouse

① 98% genetically identical.

② >20 times interbreeding between sisters & brothers.

Marking of animal:

Perminant

long time

① For small animals (mice, rabbit)

ear bunching

② For large animals (sheep, camel)

branding (burning by iron device)

temporarily

short time

① Stain

② marker

③ label on cage.

Subject

Date

No.

Bovine Serum Albumin B.S.A in phosphate buffer Saline.

mouse: 100 Mg/100ml

B.S.A : C.F.A

1 : 1

100ml : 100ml $\rightarrow$ 2ml $\Rightarrow$ I.p



20
1000
150

Jordan University of science & Technology

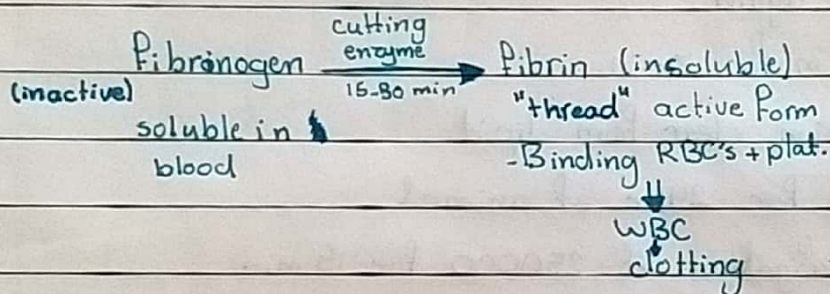
Immunology & Serology LAB

LAB(3)

Bleeding & serum Preparation

- Bleeding: cutting to take blood.

** Mechanism of clotting



Serum: collect blood in plain tube (red cap)

No anticoagulant

Type of bleeding:

- Survival (animal stay alive)
small amount of blood

① Cardiac puncture

take blood from heart

↓
heart stroke

② Vein puncture

(mice from tail) (rabbit from ear)

③ Retroorbital puncture

blood from capillary of eye.

- Termination (dead)

① Decapitation (remove of head)

"highest amount of blood"

* No need for mice

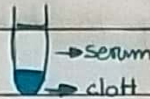
Steps for serum Preparation from mice:

- ① Bleeding (Decapitation) & clotting (coagulation)

Fibrinogen $\xrightarrow[15-30 \text{ min}]{\text{cutting}}$ Fibrin

- ② Centrifugation

at 2500 rpm for 5 min.



- ③ Transfer serum to eppendorf tube

- ④ Centrifugation again

at 2500 rpm for 3 min.

- ⑤ clearing $\rightarrow$ serum clear from lipid

- by fasting for 24 hr of animal

or - ultracentrifugation at 250000 for 15 min

- ⑥ Preservation of serum and storing.

* Preservative:

- sodium azide

- Phenol for long time

- methiolate

* Keep serum at -20°C

if I need IgE keep at -7°C

• Powder serum 4°C . (IgG)

Immunology & Serology LAB

LAB(4)

WBC (total, Blood Film)

Blood: Fluid:

- Serum: Fluid without clotting factor

• collect blood in plain tube (contain nothing) with red-orange cap.

- Plasma: Fluid with clotting factor.

• collect blood in anticoagulant tube

① No EDTA → Violet cap

② citrate

③ oxalate

} binding Ca^{+2}

④ Li heparin (green cap) → we need Ca^{+2} in the tissue culture.

Cells:

- RBC's ($4-6 \times 10^6$ cell/mm³ OR $4-6 \times 10^9$ cell/ml)

- Platelets (250000 - 400000 /mm³)

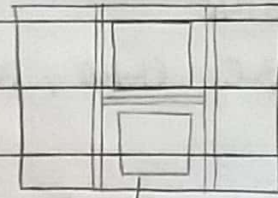
clotting

- WBC's (5000 - 12000 cell/mm³)

immune system

Total count of WBCs

Hemacytometer
 blood ← → tool
 cells

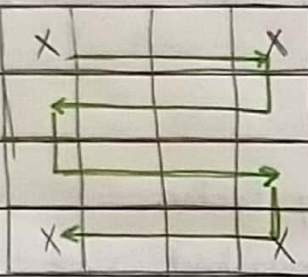


W		W
	R	
W		W

length = 1 mm

width = 1 mm

Height = 0.1 mm

V = 0.1 mm³

Volume correction factor = $\frac{1}{0.1} = 10$

$$\frac{1}{0.2} = 5$$

$$\frac{1}{0.4} = 2.5$$

W: used for count large cells

WBC's, macrophages, Fungi, ...

R: used for count small cells

RBC's, Yeast, bacteria, ...

Procedure:

- ① Take hemacytometer, clean by tap water & dry by lab coat.
- ② clean lens of microscope by lens paper.
- ③ Take 50 μ L of whole blood (EDTA tube) & Put 950 μ L acetic acid to lyse RBC's (inside eppendorf tube $\rightarrow$ mixing $\rightarrow$ brown)

$$D.F = \frac{\text{Final volume}}{\text{Sample.V}} = \frac{1000}{50} = 20$$

- ④ Put hemacytometer on microscope at 4X & Put coverslide $\rightarrow$ see lines
- ⑤ Put hemacytometer at 10X.
- ⑥ Charge (Filling) hemacytometer by Capillary tube.
- ⑦ Start count 16 smaller (1W)
- ⑧ Calculation.

$$\text{WBC's cells/mm}^3 = \text{W cells in 1W} \times D.F \times V.C.F$$

Blood Films: Thin layer

- ① differentiate between different types of WBC's
- ② see color, shape & Volume of RBC's
- ③ see platelets

Procedure:

- ① drop of blood on slide by capillary.
- ② by other slide spread blood as tongue shape.
- ③ Airdry
- ④ Add stain

{	$\rightarrow$ Wright	{	$\rightarrow$ eosin $\rightarrow$ red
	$\rightarrow$ Gimsa	{	$\rightarrow$ methelom $\rightarrow$ blue
- ⑤ wash by tap water & Airdry.
- ⑥ See under 4X, 10X, 40X

Immunology & Sereology LAB

LAB(5)

Precipitation test (D.I.D)

 $\text{Ag soluble} + \text{Ab soluble} \rightleftharpoons \text{Ag-Ab complex P.pt}$ $\text{Ag Particle} + \text{Ab soluble} \rightleftharpoons \text{Ag-Ab complex agglutination.}$

3 types of P.pt tests:

①(D.I.D): as white P.pt lines

②(S.R.I.D)

③(I.E)

D.I.D P.pt test

D: double (both Ag & Ab) are diffused

I: Immuno (Ag-Ab rxn)

D: DiFussion (movement of substance from high conc. to low conc.)

• Used for:

① Detect Purity of sample.

② Detect the homogeneity between Ag & Ab.

③ semiquantitate (Comparison)

Factors affect p.pt (agg.):

① [Ag], [Ab]

② Physical Factor (T, PH, humidity)

③ characters of Abs (affinity & valadity)

Zones of rxn (P.pt + Agg)

① Pre-zone

 $[Ag] \ll [Ab]$ non-visible rxn

② Equivalence zone

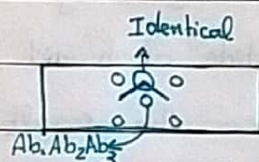
 $[Ag] \approx [Ab]$ visible rxn

③ Post-zone

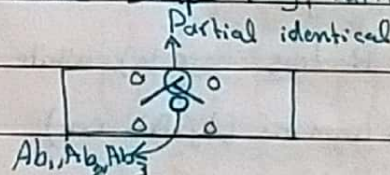
 $[Ag] \gg [Ab]$ non-visible rxn

Types of identity rxn:

① Identical

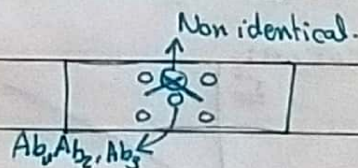
white P.pt lines are fused together (Ag. identical to Ag₂)

② Partial Identical

P.pt lines are fused with T-like shape (Ag. Partial identical to Ag₂)

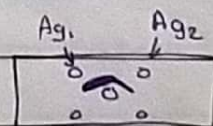
③ Non-identical rxn

two P.pt lines are crossed (Non-identical)

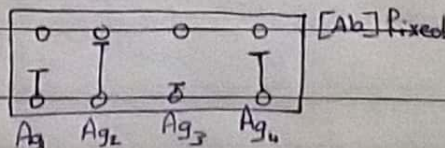


To detect semiquantitative:

① ↑ thickness of P.pt line, ↑ []

 $[Ag_1] > [Ag_2]$ 

② ↑ distance travel by substance, ↑ []

 $[Ag_2] > [Ag_1] > [Ag_3] > [Ag_4]$ 

Procedure :

- ① Pre coated slide with 1% Agarose.
- ② Add 5-6ml melting agarose 2% ($T=52^{\circ}$)
- ③ leave slide for 15-30 sec to be solid.
- ④ Put adapter on surface of agar to make wells.
- ⑤ remove agar from wells by tips.
- ⑥ Add in middle well "Anti-mouse IgG" (10M)
- ⑦ Add 10M of your Ag sample in each well from diff. mice serum.
- ⑧ Incubate slide at humid chamber at Room Temperature for 1-2 days.
- ⑨ Come to see the result.

S.R.I.D

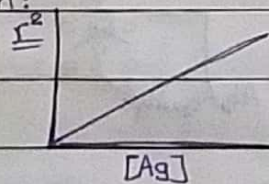
S: Single $\nearrow$ Ag or Ab (one of them) other mixed with agarose.

R: Radius (circle) while P.pt hallow

I: immuno (Ag-Ab rxn)

D: diffusion

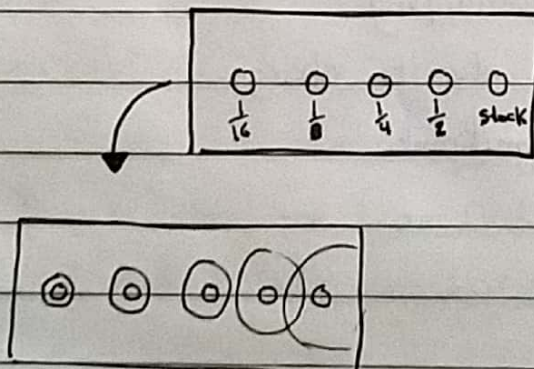
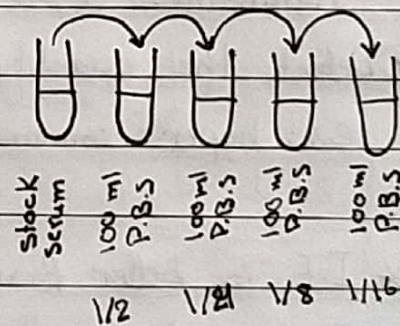
Application:



Quantitative measuring.

Procedure:

- ① Pre coating of slide with 1% agarose.
- ② mix 100 μ l "Anti mouse IgG" with 5-6 ml melting agarose (2%).
then add it to pre coating slide.
- ③ leave slide to become solid.
- ④ Make 5 wells inside agar & remove agar from well on same line by using tips.
- ⑤ Make serial dilution for serum (mice or human)
- ⑥ Take 10 μ l from each tube (diluant) & put in separate well in a sending (slide)
- ⑦ incubate slide at humid chamber at R.T for 1-2 days.
- ⑧ see the result.



Immunology & Serology LAB

LAB(6)

T.E Immuno Electrophoresis

I: immuno (Ag-Ab complex)

E: Electrophoresis

Separate of substance using electricity due to charge.

Used for:

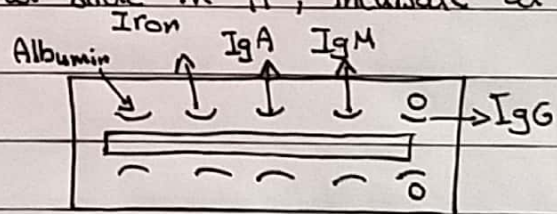
- ① Study Purity of the sample
- ② Separation of Ag (substance) due to charge.
- ③ shape & thickness of white P.pt are used for clinical significant. (বিশেষ গুরুত্ব)
- ④ Detect the quantity of immunoglobulin.
ex: hyper immunoglobulinemia (Disease)

* I.E is better than D.T.D

- ① less chance for overlapping
- ② separation of mixture due to charge.
- ③ used for clinical significant.

Procedure:

- ① Pre coating slide with 1% agarose.
- ② Add 5.6 ml melting agarose 2% at 52°C
- ③ leave slide to become solid.
- ④ make two wells in front of each others.
- ⑤ remove Agar from wells.
- ⑥ Add 10 μ L from two samples in each well. (P.S + Control)
- ⑦ Add 10 μ L loading dye in one well.
- ⑧ Put wells near -ve electrode & add tris HCl buffer in jar.
- ⑨ Run electrophoresis, $V = 120-150$, from 30 to 35 min.
- ⑩ At end of electrophoresis, Make through between two wells & remove agar from through.
- ⑪ Add "Anti human IgG" (60-100 μ L) in through
- ⑫ make humid chamber & Put slide in it, incubate at R.T for 1-2 days.
- ⑬ See your result.



Ant immune: (through)

- ① one type (homogenous)

ex. Anti human IgG-

- ② more than one type (heterogenous)

Anti human immunoglobulin (5 types)

Tris HCl $\Rightarrow$ pH = 8.6

(95%)

* most of the Protein covered by -ve charge

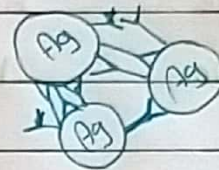
Immunology & Sereology LAB

LAB (7)

Blood Group

* Ag (soluble) + Ab (soluble) $\longrightarrow$ Ag-Ab Ppt

* Ag (particle) + Ab (soluble) $\longrightarrow$ Ag-Ab Agg
large



Direct Agg.

Ag-Ab interaction with no intermediate substances
color agg. : white sheet.

-Application

- ① Blood transfusion
- ② Detect bacterial culture
- ③ titer

	Anti A	Anti B	Anti D
AB +ve			
	P.S	P.S	P.S
	+ve	+ve	+ve

yellow : Anti B

blue : Anti A

colorless : Anti D

① ADD P.S

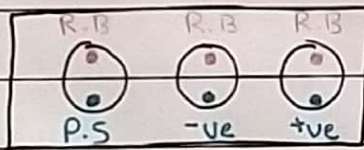
- ② Anti A, Anti B, Anti D $\longrightarrow$ mixing
- ③ see result

Blood type	Ag on surface of RBCs	Ab present in serum	suspected genotype
A $\square$	Ag A	Ab B	$I^A I^A, I^A i$
B $\square$	Ag B	Ab A	$I^B I^B, I^B i$
AB $\square$	Ag A, Ag B	—	$I^A I^B$
O $\square$	None	Ab A, Ab B	ii
$\square$ +ve	Ag Rh	None	RhRh, Rhrh
$\square$ -ve	None	None	rhrh

• Rose Bengal test (looking for Ab)

reagent Pink weakens Brucella bovis

Brucellosis: $\square$ (contaminated dairy product)



① Add 40 μ l from P.S, -ve control & +ve control
contain Ab against Brucella

② Add Rose Bengal reagent (Ag) **Pink color**

③ mix 3-5 min

④ see result.

P.S -ve $\rightarrow$ Not infected

P.S +ve $\rightarrow$ infected

Titer: Ab formed due to exposed to Ag OR highest dilution that give +ve result.

titer < 80 Previous infection

titer > 80 recent infection

titer = 80 (repeat the test after 1 month)

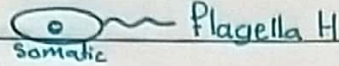
$$\frac{40}{20} = 2 \times 40$$

$$80$$

+ve	+ve	-ve	
20 ml P.S	10 ml P.S	5 ml P.S	
	$\frac{40}{10} = 4 \times 40$		
	160		
			$\frac{40}{5} = 8 \times 40$
			320

Widal test

Detect salmonella typhi & Salmonella Paratyphi (Typhoid Fever)



Salmonella typhi O → somatic → blue

Salmonella Paratyphi OA → somatic → blue

Salmonella Paratyphi OB → somatic → blue

Salmonella Paratyphi OC → somatic → blue

S. typhi H → Flagella → red

S. Para HA → Flagella → red

S. Para HB → Flagella → red

S. Para HC → Flagella → red

O	OA	S. para OB	
○	○	○	○ +ve
○	○	○	○ -ve
H	HA	HB	

① Add 40 ml P.S in all six well.

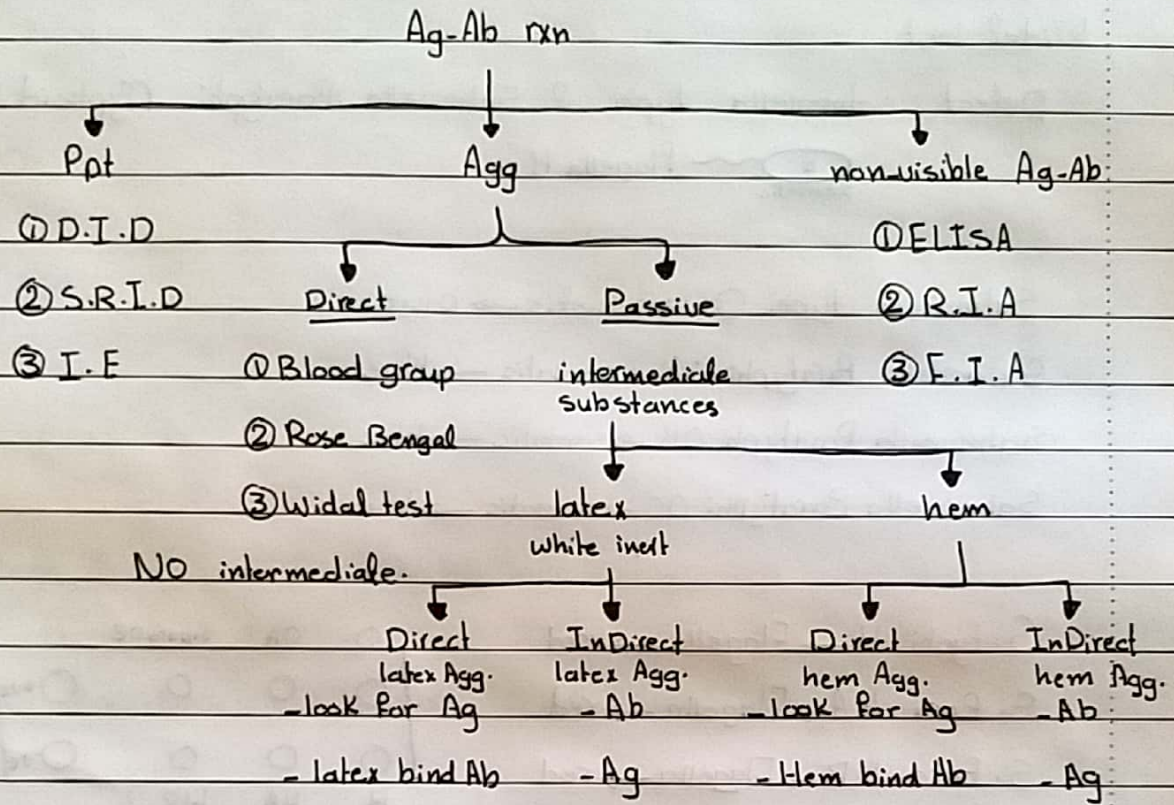
② Add Ag in well, +ve & -ve

③ mix

④ see result

+ve → titer

-ve → not infected.



Immunology & Serology LAB

LAB (8)

Direct latex agg.

- looking for Ag

- latex bind Ab

Indirect latex agg.

- looking for Ab

- looking latex bind Ag

latex: inert white substance.

• Application of Passive:

① Detect hormone.

② Detect Protein.

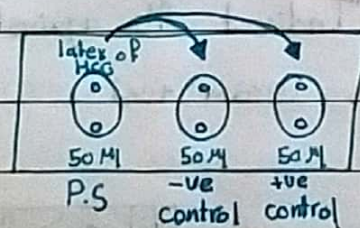
③ Detect bacterial infection.

• HCG : Ag (Direct latex Agg)

Human chorionic gonadotropic hormone.

→ Appear in blood after fertilization, in urine after One week to ten days after absence of period.

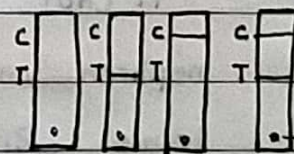
titer > 25 IU/ml



mix 3-5 min

Agg. +ve (Pregnant)

-ve (NOT Pregnant)



1 Pink line
NOT Pregnant

2 lines
Pregnant

→ 2-3 drops of urine, serum or plasma.

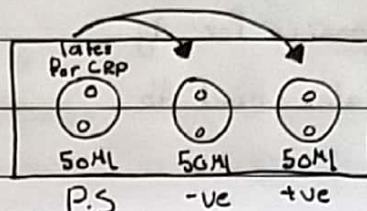
CRP: C-reactive Protein "Direct latex Agg."

Protein secreted by hepatocyte upon any inflammation.

(NOT SPECIFIC)

titer > 0.8 IU/dl

result $\leftarrow$ mix 3-5 min
+ve: infected, -ve: not infected



RF test "Indirect latex Agg."

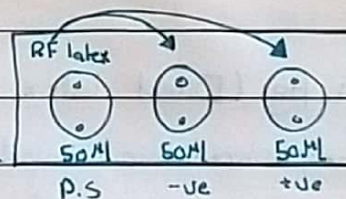
Rheumatoid Factor

Ab against Fc Portion of Ab.

elevated rheumatoid factor

titer > 8 IU/L

result $\leftarrow$ mix 3-5 min
+ve agg.: R. arthritis.
-ve agg.: not infected.



ASO: Antistreptolysin O "Indirect Ab against E"

β -hemolytic Streptococcus (Gram +ve)

$\uparrow\uparrow$ [Ab]

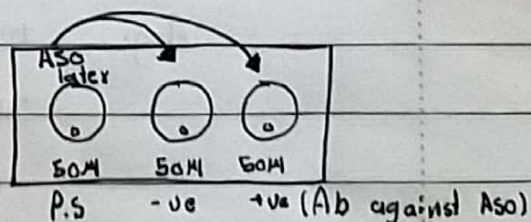
nephron $\rightarrow$ nephritis

joints $\rightarrow$ R. arthritis

Cardiac $\rightarrow$ endocarditis

result $\leftarrow$ mix
+ve: infected -ve: not infected.

titer > 200 IU/ml



Immunology & Sereology LAB

LAB (9)

ELISA

E: Enzyme

L: linked

I: immuno (Ag-Ab)

S: Sorbant (adsorp to plastic surface)

A: Assay $\rightarrow$ $\uparrow$ Product/min

$\downarrow$ substrate/min

① Indirect ELISA : looking For Ab.

② Direct ELISA : looking For Ag. (Sandwich method ELISA)

③ Competitive ELISA $\approx$ R.I.A

- 3 Types of bottom of well:

① V-bottom

③ U Flat bottom

② U-bottom

Used For

① Turner marker

② Determine infectious disease

③ Some Proteins & hormones

Hb Ag, Rubella, AID

Advantage of ELISA:

① simple.

② safe.

③ No need for expensive tool.

④ Reliable.

Result: ELISA reader

specific type of spectrophotometer.

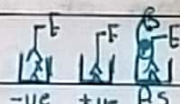
Enzyme used in ELISA:

- ① High turnover number.
- ② React with chromogen substances
(substrate colorless, Product color)
- ③ E can bind Abs (HRP- Antihuman IgG)
- ④ Specific & stable.

E	substrate	Product
A.P. (Alkaline phosphatase)	PNP: P (colorless)	PNP (yellow)
Glucosidase	λ Glu (colorless)	Glu (blue)
H.R.P. (Horse radish) Peroxidase	ABTS	→ green
	O.P.D	→ orange
	T.M.B tetra	→ blue ↓ stop yellow

Indirect ELISA

Hb₂ Ag



- ① Plate (wells) coated with Ag (Hb₂ Ag) 50-150 TU/ml
- ② washing & Add blocking substance to bind non specific site - washing
- ③ Add 50 ml -ve, 50 ml +ve, 50 ml P.S → incubate then washing
- ④ Add E-Ab₂ (Conjugate) 50 ml
(50-70 min)
- ⑤ Incubate plate for 1 hour at 37°C
- ⑥ washing 5X (350 ml - 400 ml) / well
- ⑦ Add substrate $\left[\begin{array}{l} \text{ch. A} \rightarrow \text{drop} \\ \text{ch. B} \rightarrow \text{drop} \end{array} \right]$ incubate 15 min at 37°C.
- ⑧ No washing, Add stop solution drop (blue → yellow)
- ⑨ → color ↑ color ↑ [S] ↑ [Ab]

at $\lambda = 450 \text{ nm}$

Direct ELISA: (sandwich)

- ① Plate coated with Ab₁
- ② Add blocking at nonspecific site
- ③ 50 ml -ve, 50 ml +ve, 50 ml P.S, ... washing
- ④ Add E-Ab₂ → incubate 1 hr at 37°C.
- ⑤ washing to remove non-binding ~~substrate~~ substance 5X.
- ⑥ Add substrate $\begin{bmatrix} A \\ B \end{bmatrix}$ incubate 15 min at 37°C.
- ⑦ No washing. Add stop solution
- ⑧ Result → color ELISA $\lambda = 450$
eye

↑ color ↑ [S] ↑ [Ag]

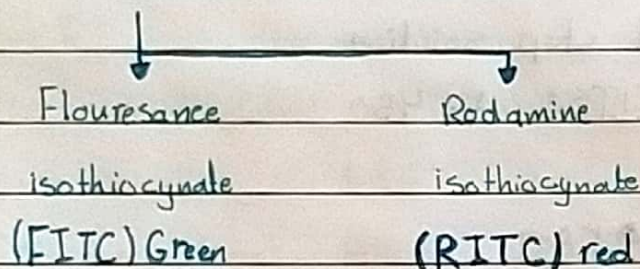
Immunology & Serology LAB

LAB (10)

Flourescent Immuno Assay F.I.A

Flourescent: salt, that have colorless, absorb U.V, converted into visible light.

Flourescent salt (Slides)



Used For:

- ① Detect histochemical substance (inside tissues) → element, drug.
- ② Detect some infection
 - ① Bacterial intracellular (trepanoma Pallidium (Syphilis))
 - ② Parasitic toxoplasma gonidi (Toxoplasmosis)
 - ③ Respiratory Cynthal virus R.C.V → respiratory infection.
- ③ Detect autoimmune disease
ex: Systemic lopus Erythramatosis, autoimmune heptitious, autoimmune hemolytic anemia, Rematoid Arthritis.

Qualitative

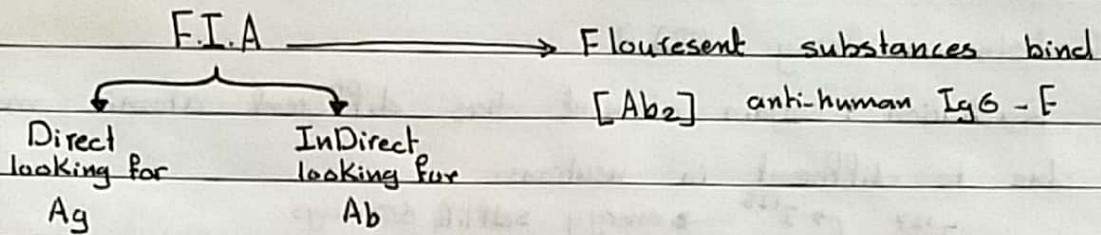
+ve → Flouresence

-ve → not Flouresence

eye → Flouresent.

Quantitative

Attached Flouresent microscope to photometer → reading
Fluorometer



ANA : Antinuclear Antibody

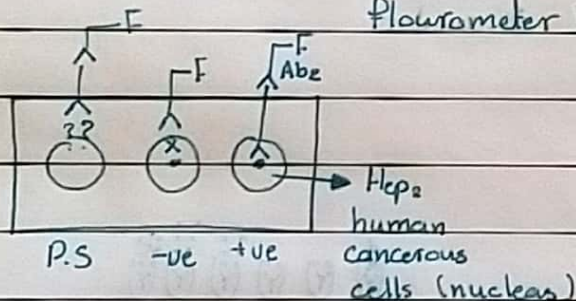
titer normally < 40

> 40 : have antinuclear disease

ANA → Indirect F.I.A

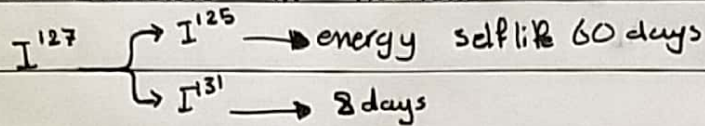
- ① slide coated by Hep₂ cells (nucleus)
- ② P.S., -ve control, +ve control, washing
- ③ Add F-Ab₂ to all → incubate → wash
- ④ See result under flourescent microscope by eye.

Fluorometer by reading.



Radioimmuno Assay (R.I.A)

Radiation : certain element has different atomic mass due to different in neutrons.



Used for detect Ag:

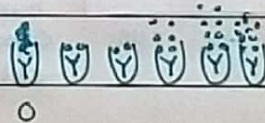
- ① hormones T_3, T_4
- ② some drug cortisolone
- ③ Tumor marker (low)

very sensitive specific for sensitive Pg

Disadvantage:

- ① Dangerous $\rightarrow$ mutation
- ② Costly $\rightarrow$ instrument γ -counter.
- ③ specific Precaution
- ④ Time.

• Procedure : (hormone)



$T_4: 60-160 \text{ M.M/L}$

- ① test tubes coated with fixed conc. of Ab against T_4
- ② Add cold Ag (bottles contain conc. of AgT_4) to make standard. (not have radiation)
- ③ Add P.S.1, P.S.2 ..., incubation cold Ag: P.S curve
- ④ Add hot Ag (fixed conc.) $\approx 50000 \text{ cpm}$
- ⑤ incubation $\rightarrow$ remove excess hot Ag. $\rightarrow \gamma$ -counter curve

$T = \text{total}$

$B = \gamma \text{ counter}$

$P_{S1} = 80$

$P_{S2} = 40$

